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PREDICT REPORT

The purpose of this project is to provide a functional remote satellite tracker that indicates the specific time windows during which the satellite can be seen with respect to the observer's position on Earth. The factors considered in this project not only indicate satellite visibility but also utilise orbit prediction by identifying the primary COEs (Classical Orbital Elements) and examining how they are affected by orbital perturbations.

This satellite tracking system version is limited to the observer's latitude in the range from 0° to +90°, meaning it only functions for the northern hemisphere. This tracking system demonstrates a clear example of applying astrodynamic theory to real-world scenarios, and it encompasses the basics of orbital visualisation and satellite prediction.

Before diving into the system's functionality, assumptions have to be set for simplifying the mathematics and the core concepts to execute orbit prediction and orbit functionality. The Earth is assumed to be spherical, bulging in the centre, and its drag affects some orbital elements. The Earth’s shadow is cylindrical. The Earth is half light and half dark, meaning that there is an instantaneous sunrise and sunset. All the satellites' orbits are assumed to be ellipses. The time used to calculate the predictions is in UT, which is then transformed into Julian days to facilitate mathematical calculations. The weather conditions for the satellite sightings must be clear and dark. Other pull forces from third bodies are not taken into account; only the EOM (equation of motion) assumptions are used. Fuel and velocity of the satellites are not being considered.

As mentioned above, the theory behind this system relies on updating the COE data collected from the data provided and converting that to ρ, Az and El parameters. Starting with the Echo data supplied by the user. These factors are site latitude, site longitude, site altitude, current year, a start date, and an end date, and an end date to end the satellite's tracking. This data is transformed into working units of radians and Julian days. From the initial orbital elements provided, the system calculates the range of time and how the J2 effect and the drag affect the orbital elements. These are used to determine the rate of change of the right ascension node (**Ω**), argument of perapsis (**ω**), and the eccentricity (*e*). The equations are provided in Appendix A.

Then, to predict the updated COEs, Taylor series were used. For slow-changing COEs, the first-order Taylor series was used. For fast-changing COEs, the second-order Taylor series was used. Now, after calculating the updated COEs for every time step between the start time and the end time, the position of the satellite in the ECI coordinate system can be determined. Then the transformation from the ECI to the ECEF (topocentric) coordinate system has to be made to match the satellite's movement with the observer's movement. Subtracting the distance from the satellite from the distance from the observer. A rotation subtracting 90° from the site's altitude with respect to the y-axis, multiplied by the rotation of the local sidereal time with respect to the z-axis. This is called the slant range vector from which the system extracts the site's final ρ Az, and El parameters. The respective equations are shown in Appendix B.

Once all the necessary values are calculated, the system can finally determine if the satellite is visible with respect to the observer's position. The system's validation process includes many

factors to output the time span and visibility of the satellite. The four preliminary checks for a satellite's visibility are as follows: The satellite's elevation (El) must be above the horizon (El > 10°). The observing site must be in darkness, which is determined by the angle alpha (α)—the angle between the Sun's vector and the observer's vector relative to the ecliptic plane; alpha must be greater than 90°, meaning the observer is in darkness. Additionally, to ensure the satellite itself is illuminated, the angle beta (β)—the angle between the Sun's vector and the satellite's vector with respect to the ecliptic plane—must indicate the satellite is in sunlight and not in Earth's shadow. Finally, the distance from the observer to the satellite should be less than 1500 km for naked-eye visibility. The corresponding calculations for the angles alpha and beta are shown in Appendix C.

The system will only output the times when the site is dark, and it will indicate if the satellites can be seen or not. Even if the satellite can't be seen, the system will output the satellite's ρ Az, and El parameters. The final data that was provided to the program has around 100 satellites, so there are different time frames for satellites to be seen without the use of a telescope. The intended results are shown in Appendix D.

In order to achieve this, the system was only tested with one satellite's data. The main challenges that I ran into while developing the system's code were managing the time references properly. The system loops through two-minute time steps. In the first trials, the system was running uneven time steps rather than even times as it should. It was also challenging to arrange all of the different validation conditions. I had to ensure that the conditions did not overlap and that the code executed them in the correct order. The validation sequence is as follows: First, check if the site is in darkness. If it is, calculate the ρ, Az, and El parameters. If not, the system proceeds to

the next time step and skips the other conditions. Then, if ρ and El meet the previously mentioned criteria, the program finally displays whether the satellite is visible: “YES” or “NO.” A recommendation for improving the system accuracy would be to include a southern hemisphere sunlight model by implementing other assumptions regarding the Earth’s shadow and generalising the geodetic-to-ECEF and topocentric conversion rotations accordingly to correctly handle all latitudes (–90° to +90°).

To operate the program, the user inputs the required satellite and observer information, such as the site latitude, longitude, altitude, the current year, and the desired start and end dates, into the MATLAB script, which then processes all values in radians, kilometres, seconds, and Julian days. The respective program input display is shown in Appendix E. Once executed, the program automatically updates the orbital elements, propagates the orbit through the selected time span, and outputs the slant range (ρ), azimuth (Az), and elevation (El), along with the satellite’s visibility status. The system is limited to observer latitudes between 0° and +90°, meaning it only functions for the northern hemisphere, and it assumes all orbits are elliptical with simplified perturbation models. Additionally, because the visibility model depends on darkness conditions and assumes a cylindrical Earth shadow, the program may not correctly handle extreme latitudes, non-elliptical trajectories, or satellites requiring higher-order perturbation modelling. A unique aspect of the program is its integration of both J2 and drag perturbations with mixed-order Taylor series propagation, allowing efficient computation over long time intervals. However, the program will not work properly if provided incomplete orbital data, unrealistic time ranges, or geodetic coordinates outside its defined limits.

Appendix A. J2 Effect and Drag Perturbation Calculations:

$\mu = 398600.5\ km^3/sec^2$, $R \oplus = 6378.137 km$, $J_2 = 1.082663 \cdot 10^{-3}$

$$\overline{\dot{n}} = n_0 \left[1 + \frac{3}{2} J_2 \left(\frac{R_\oplus}{p_0} \right)^2 \sqrt{1 - e_0^2} \left(1 - \frac{3}{2} \sin^2 i_0 \right) \right]$$

$$n = n_0 + \dot{n}_0 \Delta t$$

$$\dot{e}_{drag} = \frac{-2(1 - e_0)\dot{n}_0}{3\overline{n}}$$

$$e = e_0 + \dot{e}_{drag} \Delta t$$

$$\dot{\Omega}_{J_2} = \left[-\frac{3}{2} J_2 \left(\frac{R_\oplus}{p_0} \right)^2 \cos i_0 \right] \overline{n}$$

$$\Omega = \Omega_0 + \dot{\Omega}_{J_2} \Delta t$$

$$\dot{\omega}_{J_2} = \left[\frac{3}{2} J_2 \left(\frac{R_\oplus}{p_0} \right)^2 \left(2 - \frac{5}{2} \sin^2 i_0 \right) \right] \overline{n}$$

$$\omega = \omega_0 + \dot{\omega}_{J_2} \Delta t$$

Appendix B. Preliminary Orbit Determination:

$$\begin{bmatrix} \mathbf{I} \\ \mathbf{J} \\ \mathbf{K} \end{bmatrix} = [\mathbf{Rot3(-LST)}][\mathbf{Rot2(-COLAT)}] \begin{bmatrix} \mathbf{S} \\ \mathbf{E} \\ \mathbf{Z} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{ROT2} \\ \beta \end{bmatrix} = \begin{bmatrix} \mathbf{cos}\beta & \mathbf{0} & -\mathbf{sin}\beta \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{sin}\beta & \mathbf{0} & \mathbf{cos}\beta \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{ROT3} \\ \gamma \end{bmatrix} = \begin{bmatrix} \mathbf{cos}\gamma & \mathbf{sin}\gamma & \mathbf{0} \\ -\mathbf{sin}\gamma & \mathbf{cos}\gamma & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

Vector Magnitude (Range)

$$\rho = \|\mathbf{R}_{SEZ}\|$$

Elevation Angle

$$\text{el} = \arcsin\left(\frac{R_{SEZ,z}}{\rho}\right)$$

Azimuth Angle

$$\text{az} = \arccos\left(\frac{-R_{SEZ,x}}{\rho \cos(\text{el})}\right)$$

Appendix C. Validation Calculations:

Angle α (Sun–Site Angle)

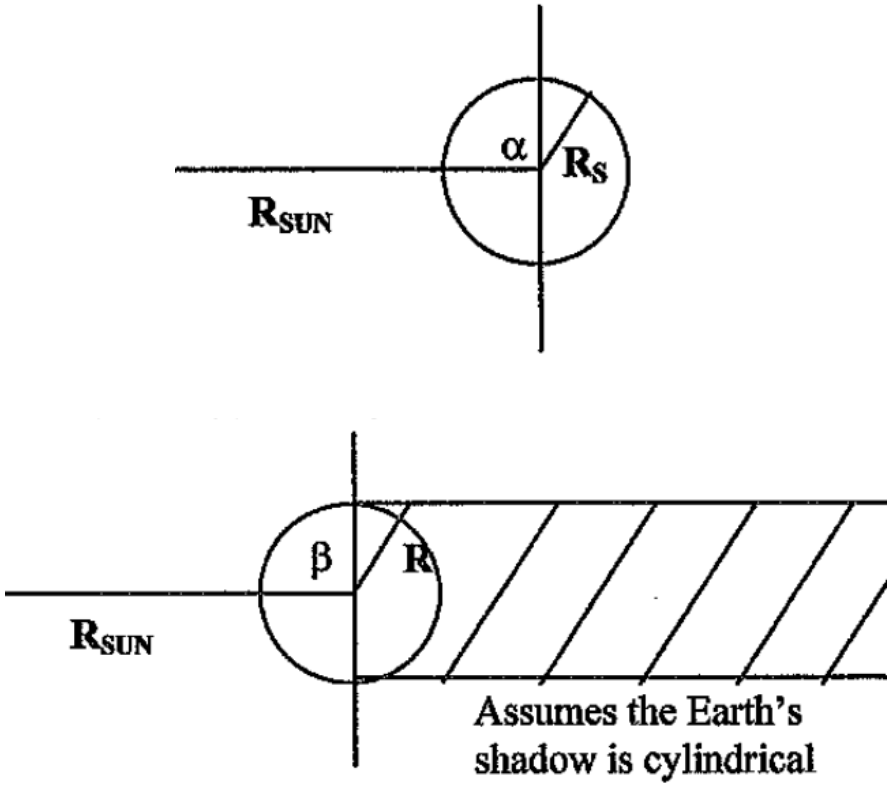
$$\alpha = \frac{\mathbf{R}_{Sun} \cdot \mathbf{R}_{site}}{\|\mathbf{R}_{Sun}\| \, \|\mathbf{R}_{site}\|}$$

$$\alpha = \arccos(\alpha)$$

Angle β (Sun–Satellite Angle)

$$\beta = \frac{\mathbf{R}_{Sun} \cdot \mathbf{R}_{ijk}}{\|\mathbf{R}_{Sun}\| \, \|\mathbf{R}_{ijk}\|}$$

$$\beta = \arccos(\beta)$$



Appendix D. Tracking System Output:

```
 Echo Check Input Data:

     Site Latitude  =     38.880 deg
     Site Longitude =   -108.560 deg
     Site Altitude  =      1.515 km

     Year           =       2025
     Start Day      =      320.0
     End Day        =      327.0

 Echo Check Working Units

     Site Latitude  =  0.67858401 rad
     Site Longitude = -1.89472944 rad

     Start Julian Day = 2460995.50000
     End Julian Day   = 2461002.50000

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 Echo Check Input Data for: ASTEX 1

    Epoch Year     =         2025
    Epoch Day      = 320.69926039
    N Dot/2        =   0.00001265  rev/day^2
    Inclination    =  92.71220000  deg
    RAAN           = 115.19690000  deg
    Eccentricity   =   0.00178010
    Arg of Perigee = 104.34660000  deg
    Mean Anomaly   = 255.97180000  deg
    Mean Motion    =  14.50597454  rev/day
```

```
 Echo Check Working Units

    Epoch Time (month/day, hr:min:sec) =  11/16,  16:46:56.10

    Epoch Julian Day (days) =  2460996.19926039

    N    = 0.00105490424 rad/s Ndot2     =  +2.12947675904e-14 rad/s^2
    Ecc  = 0.00178010000       Ecc dot   =  -1.34424193971e-11 1/s
    Inc  = 1.61813314677 rad   Mean Anom =  +4.46755070226e+00 rad
    RAAN = 2.01056519309 rad   RAAN dot  =  +6.53834356549e-08 rad/s
    Argp = 1.82119173326 rad   Argp dot  =  -6.83142490317e-07 rad/s

MONTH/DAY   HR:MIN (UT)   RHO(KM)     AZ(DEG)    EL(DEG)   VIS?
-------------------------------------------------------------------
    11/16   10:19    1237.086      92.920    31.522   NO
    11/16   10:21    1337.594      45.724    27.990   NO
    11/17   00:01    1266.746     300.512    30.172   YES
    11/17   00:03    1347.193     254.366    27.323   YES
    11/17   11:07    1210.622     188.163    32.450   NO
    11/17   11:10     764.960     243.505    70.252   NO
    11/17   11:12    1078.549     338.776    38.812   NO
    11/18   10:19    1224.614     110.628    31.903   NO
    11/18   10:21    1147.387      56.570    35.294   NO
    11/19   00:01    1451.005     310.815    24.382   YES
    11/19   00:03    1371.791     269.512    26.630   YES
    11/19   11:07    1459.892     191.977    24.068   NO
    11/19   11:10     887.263     220.787    52.318   NO
    11/19   11:12     947.237     319.596    47.202   NO
    11/20   10:19    1278.161     127.338    29.775   NO
    11/20   10:21    1005.955      73.329    42.914   NO
    11/20   10:23    1363.977      25.759    27.034   NO
    11/21   00:03    1452.822     282.993    24.336   YES
    11/21   11:10    1082.651     214.735    38.241   NO
    11/21   11:12     905.482     290.586    50.565   NO
    11/21   11:14    1385.020     332.475    26.355   NO
    11/22   10:19    1388.776     141.231    26.041   NO
    11/22   10:21     934.266      98.122    47.980   NO
    11/22   10:23    1134.569      29.721    35.773   NO
```

Appendix D. Tracking System Input:

```
ENTER THE SITE LATITUDE IN DEG  38.880
ENTER THE SITE EAST LONGITUDE IN DEG  -108.560
ENTER THE SITE ALTITUDE IN KM  1.515

ENTER THE START DAY OF YEAR  335
ENTER THE STOP DAY OF YEAR  339
ENTER THE YEAR OF OBSERVATION (IE 1997, 2000, ETC)  2025
ATLAS CENTAUR 2
```